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ALLOCATING FLUID FLOW BY CONTROLLING INTELLIGENT COMPLETION VALVES IN A HYDROCARBON WELL USING MACHINE LEARNING

Abstract

A system for controlling intelligent completion valves (ICVs) used in a hydrocarbon well operation is disclosed. The ICVs can be located downhole in a wellbore, and an outflow of pressurized injection fluid from the wellbore may be used to drive hydrocarbons toward one or more offset producing wells. A trained machine-learning model can be generated by training a machine-learning model on training data comprising acoustic signal data generated by a pressurized injection fluid flowing through the ICVs at various flow rates and ICV positions. When applied to new acoustic sensing system sensor data associated with an ICV of multiple ICVs in the well, the trained machine-learning model can generate a result indicating a predicted flow rate of the pressurized injection fluid through the ICV. The result may be output and used to control the flow rate of pressurized injection fluid through the ICVs.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to wellbore operations, and more particularly (although not necessarily exclusively) to controlling intelligent completion valves to allocate fluid flow from a well using machine learning.

BACKGROUND

[0002] A hydrocarbon well may be designed with a target flow rate to generate a desired rate of hydrocarbon production at one or more offset wells without inducing pressures that might harm the formation within which the wells are located. Such wells may utilize intelligent completions, which can allow the well operator to control downhole processes from the surface. For example, an intelligent completion may include control valves, sensors, or other devices via which downhole production or injection processes can be controlled and monitored from the surface. Physics-based models are commonly used to estimate control valve settings that will produce the desired target flow rates, and physics-based models typically rely on reservoir properties for this purpose. But reservoir properties can change and thus an actual flow rate of a well may resultantly deviate from a target flow rate. It may be difficult for an operator located at a well surface to detect such a flow rate deviation in a timely manner, and likewise difficult to attribute a flow rate deviation to a particular control valve(s). As a result, the rate of hydrocarbon production at the offset well(s) may differ from an intended or expected production rate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 is a schematic diagram of a hydrocarbon well system according to one example of the present disclosure.

[0004] FIG. 2 is a block diagram depicting one example of a computing device for controlling downhole intelligent completion valves, according to an example of the present disclosure.

[0005] FIG. 3 is a block diagram of an application for training a machine-learning model to generate a result that can be used to control downhole intelligent completion valves, according to an example of the present disclosure.

[0006] FIG. 4 is a block diagram of a machine-learning-based system including the computing device of FIG. 2. for controlling downhole intelligent completion valves, according to an example of the present disclosure.

[0007] FIG. 5 is a flow chart representing one example of a method for controlling downhole intelligent completion valves according to an example of the present disclosure.

DETAILED DESCRIPTION

[0008] Certain aspects and examples of the present disclosure relate to a machine-learning-based system for controlling intelligent completion valves (ICVs) in a hydrocarbon well to allocate fluid flow from the well. In one example, the ICVs can be located in a completed injection well that is drilled in a formation and into which is directed a pressurized injection fluid. The pressurized injection fluid can be emitted from the injection well and injected into the formation through the ICVs and corresponding perforations in the injection well. The flow rate of pressurized injection fluid into the formation may be controlled, at least in part, by the ICVs, which can allow target flow rates through the formation to be maintained.

[0009] The flow of pressurized injection fluid through the ICVs produces resulting acoustic signals. An acoustic sensing system may be used to record the acoustic signals produced by the ICVs. The flow rate of the pressurized injection fluid within the injection well can also be measured in the

area of each ICV and recorded concurrently with recording by the acoustic sensing system of the acoustic signals generated by the ICVs. The recording of the flow rates and the acoustic signals may occur at different ICV positions to correlate pressurized injection fluid flow rate data and ICV acoustic signal data with ICV position data. Likewise, the recording of acoustic signals may occur at different pressurized injection surface fluid flow rates to correlate acoustic signal data with pressurized injection fluid flow rate data. The recording of the flow rates and the acoustic signals may also occur at different pressurized injection fluid pressures. In some examples, the flow rate of the pressurized injection fluid within the injection well can be measured and recorded using an injection logging tool.

[0010] In some examples, acoustic signal data and pressurized injection flow rate data may be obtained from a well other than a well of current interest. In other examples, the training data may be data obtained directly from the well of current interest, such as by operating an injection logging tool in the well of current interest to record pressurized injection fluid flow rates while simultaneously recording ICV acoustic signal data using an acoustic sensing system.

[0011] Recorded acoustic signal data and corresponding recorded pressurized injection fluid flow rate data can serve as training data for training a machine-learning model. More specifically, a trained machine-learning model can be generated by training the machine-learning model on training data that includes the recorded acoustic signal data generated by the flow of pressurized injection fluid through the ICVs at various recorded flow rates of the pressurized injection fluid, and at different ICV positions and possibly different pressurized injection fluid pressures. Other data, such as various engineered feature data, may also be part of the training data in some examples.

[0012] New sensor data received by the distributed acoustic sensing system for a given ICV can thereafter be applied to the trained machine-learning model. Based on the input of the new sensor data, the trained machine-learning model can generate a result indicating a predicted flow rate of pressurized injection fluid through the ICV.

[0013] In some examples, the system may use the result generated by the trained machine-learning model to control the associated ICVs in a well. For example, the system may adjust a position of one or more of the ICVs to maintain a target flow rate based on predicted ICV flow rates output by the trained machine-learning model. In other examples, the system may provide the predicted flow rates output by the trained machine-learning model to another control system, which can use the predicted flow rates to control the associated ICVs in a similar manner. In some examples, the flow rate of the pressurized injection fluid through the ICVs in a well may be controlled by adjusting the pressure of the pressurized injection fluid delivered to the well based on predicted ICV flow rates output by the trained machine-learning model.

[0014] Controlling ICVs can be used to increase efficiency of the wellbore operation. For example, an outflow rate of pressurized injection fluid from an injection well into a formation that is too high may result in water breakthrough, where hydrocarbons in the formation are displaced by water that can migrate to the producing well(s) and reduce the amount of hydrocarbons extracted from the formation. Injecting too much injection fluid into the formation may also cause more severe issues such as a loss of caprock integrity, which can lead to problems with hydrocarbon envelope containment. Contrarily, an outflow rate of pressurized injection fluid from an injection well into a formation that is too low may result in a hydrocarbon production rate at one or more offset wells that is less than expected. Controlling ICVs of a hydrocarbon well operation may allow target flow rates to be more easily achieved and maintained, which can allow the well field to be more accurately managed and can improve production and sweep efficiency.

[0015] Illustrative examples follow and are given to introduce the reader to the general subject matter discussed herein rather than to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the

illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

[0016] One example of a hydrocarbon well field **100** is represented in the diagram of FIG. **1**. As shown, the hydrocarbon well field includes a completed injection well **102** and at least one completed offset producing well **150** located in a reservoir. The injection well **102** can include a wellbore **104** that is drilled in a subterranean formation **106** within the reservoir, but may alternatively be drilled in a sub-oceanic formation in other examples. In this example, the injection well **102** includes only a vertical wellbore **104**. In other examples, the injection well **102** may also have a horizontal wellbore portion. The injection well **102** may include a wellbore casing **108**, which may be cemented into the wellbore **104** by introducing cement **110** into an annular space between the wellbore **104** and wellbore casing **108**.

[0017] The injection well **102** may include multiple sets of perforation clusters **112** that are disposed at intervals along the wellbore **104** and pass through the wellbore casing **108** and the cement **110**. In this example, a set of perforation clusters **112** coincides with each of multiple treatment stages of the injection well **102**, with each treatment stage being associated with a different zone of the formation **106**. The various treatment stages of the injection well **102** may be separated using plugs or other suitable elements to isolate stages, and each treatment stage may have more than one set of perforation clusters.

[0018] In the case of the completed injection well **102**, the perforation clusters **112** allow pressurized injection fluid **114** to be emitted from the wellbore **104** and injected into the formation **106** as indicated. In this example, the injection fluid is obtained from an injection fluid source **116** (e.g., a tank) and pressurized using one or more pumps **118** before being delivered into the wellbore casing **108** of the wellbore **104**. The pump(s) **118** may be a fixed position pump, one or more mobile pumps such as pump trucks, etc. As can be observed in FIG. **1**, the pressurized injection fluid **114** will flow forcibly outward through the perforation clusters **112** and into the formation **106**. The flow of pressurized injection fluid **114** into the formation **106** can encourage hydrocarbons located in the formation **106** between the injection well **102** and the producing well **150** to migrate toward the producing well **150** for extraction, as indicated by the arrows **120**. In some examples, the injection well **102** may be a water injection well.

[0019] The outflow of the pressurized injection fluid **114** from the injection well **102** into the formation **106** can be controlled by a plurality of intelligent completion valves (ICVs) **122** installed downhole in the injection well **102**. The ICVs **122** may be part of an intelligent completion, which may also include sensors or other devices that can facilitate monitoring and control over the injection well **102**. In at least some examples, a position (i.e., open, closed, partially open) of the ICVs **122** can be adjusted from the well surface **124**. For example, the position of an ICV **122** may be adjusted via hydraulic fluid delivered from the well surface **124** through a hydraulic supply line. A position adjustment of an ICV **122** may be initiated by an operator, a computing device **200**, or a separate process (ICV) control device **250** located at the well surface **124**.

[0020] In the example depicted in FIG. **1**, the ICVs **122** in the injection well **102** may be part of a tubing-conveyed completion that is located in the wellbore **104**. The tubing-conveyed completion may include a tubing string **126**, such as a coiled tubing string. Multiple isolation elements **128** may be coupled to the tubing string **126** along the length thereof, and may be located to straddle the various perforation clusters **112** as shown. The isolation elements **128** can be used to isolate different zones and corresponding perforation clusters **112** within the wellbore **104**. In this regard, the isolation elements may be deployed in one or more pairs, where at least one first isolation element is positioned on the uphole side and at least one cooperating isolation element is positioned on the downhole side, of each set of the perforation clusters **112** in the wellbore **104**. For example, as shown in FIG. **1**, the various pairs of isolation elements **128** may be used to isolate four separate zones, denoted as Zone **1** through Zone **4**, where each isolation zone encompasses a perforation cluster **112**. The isolation elements **128** may be individually controllable or may be controllable in

cooperating pairs, to isolate an individual zone of the wellbore **104** around a given set of the perforation clusters **112** or to simultaneously isolate multiple zones of the wellbore **104** around several sets of the perforation clusters **112**. In such an example, one or more ICVs **122** may be located on the tubing string **126** between each cooperating pair of isolation elements **128**. The ICVs **122** may, but are not required to be, positioned to be substantially aligned with the perforation clusters **112** bounded by the isolation elements **128**, to facilitate passing of the pressurized injection fluid **114** emitted by the ICVs **122** through the perforation clusters **112** and into the formation **106**.

[0021] By employing a tubing-conveyed completion having multiple isolation elements **128** as described, the flow of pressurized fluid may be estimated or allocated on a per ICV basis, a per isolation zone basis, or a per cluster basis. In some examples, the isolation elements **128** may be inflatable packers, the inflation and deflation of which may be controllable from the well surface **124**. In some examples, the isolation elements **128** may be swell packers that can expand once in contact with wellbore fluids such as brine, water, or hydrocarbons. While the example of FIG. **1** depicts the injection well **102** as having four isolation zones with one perforation cluster **112** per zone shown, it should be understood that a different number of isolation zones and a different number of perforation clusters **112**, including a different number of perforation clusters per isolation zone, may be present in other well examples.

[0022] As with the injection well **102**, the producing well **150** may include a wellbore **152** that is drilled in the subterranean formation **106**. In this example, the producing well **150** includes only a vertical wellbore **152**. In other examples, the producing well **150** may also have a horizontal wellbore portion. The producing well **150** may include a wellbore casing **154** for transporting hydrocarbons **156** to the well surface **124**. The wellbore casing **154** of the producing well **150** may also have cement **158** located between the wellbore casing **154** and the wellbore **152**. The completion of the producing well **150** may include a wellhead **162** (e.g., Christmas tree) or another well completion apparatus such as a pump, a derrick, etc.

[0023] The producing well **150** may also include multiple sets of perforation clusters **160** that are disposed at intervals along the wellbore **152** and pass through the wellbore casing **154** and the cement **158**. The sets of perforation clusters **160** may again coincide with each of the multiple treatment stages of the injection well **102** and with the different zones of the formation **106**. In the case of the producing well **150**, the perforation clusters **160** can allow the hydrocarbons from the formation **106** to enter the producing well **150** and to be transported to the well surface **124**.

[0024] The outward flow of the pressurized injection fluid **114** through the perforation clusters **112** and the ICVs **122** of the injection well **102** creates acoustic emissions at the ICVs, the properties of which can be captured as acoustic signals. The particular sound and intensity (amplitude) of a given acoustic emission is related to the position of the ICV **122** through which the pressurized injection fluid **114** passes. For example, the amplitude of an acoustic signal generated by the outward flow of the pressurized injection fluid **114** through an ICV **122** and an associated set of perforation clusters **112** may increase as the degree to which the ICV **122** is opened increases.

[0025] The acoustic signal associated with the flow of the pressurized injection fluid **114** through the perforation clusters **112** and the ICVs **122** of the injection well **102** can be recorded for each ICV **122** using an acoustic sensing system **130**. In some examples, the acoustic sensing system **130** can collect DAS data, DTS data, downhole pressure gauge data, surface pressure gauge data, or surface injection flow data. The recorded differential phase (acoustic) DAS data may be a 2-dimensional measurement of strain rate with respect to measured depth and time.

[0026] In the example of FIG. **1**, the acoustic sensing system **130** is a distributed acoustic sensing (DAS) system. The DAS system can include an optical fiber cable **132** that is located in the wellbore **104** of the injection well **102**. In some examples, the optical fiber cable **132** can be clamped to or cemented to the outside of the wellbore casing **108**. In other examples, the optical fiber cable **132** can be deployed into the casing **108** of the injection well **102** along with a wireline, a slickline, or a permanent cable, may be deployed along with and clamped or otherwise affixed to

the tubing string **126** as shown, or may be deployed by another suitable technique. The optical fiber cable **132** may also be deployed into the injection well **102** by gravity, pumping, or by pushing or otherwise inserting the optical fiber cable **132** into the injection well **102**. The optical fiber cable **132** may also be self-propelled using mechanical or chemical propulsion. The optical fiber cable **132** can be various types of optical fiber cables. For example, the optical fiber cable **132** may be a lower cost disposable optical fiber cable.

[0027] The optical fiber cable **132** extends, in this example, from a wellhead exit of the wellbore casing **108**, along the surface **124** of the formation **106**. A surface optical fiber cable **134** can connect the optical fiber cable **132** attached to the completion tubing string **126** with an optoelectronic interrogator of the DAS system. The optical fiber cable **132** and the optoelectronic interrogator may form at least a primary portion of the DAS system. The DAS system may utilize various implementations of, for example, Rayleigh or Brillouin scattering, and may be interferometric in nature. Employed sensing technologies may include, for example, homodyne, heterodyne, Michelson, Mach-Zender, Fabry-Perot, phase based, intensity based, coherence based, static (absolute) or dynamic (relative) sensing principles.

[0028] The acoustic sensing system **130** may be communicatively coupled to a computing device **200**. The computing device **200** can receive acoustic signal data from the acoustic sensing system **130** and may use the acoustic signal data within the process of controlling the ICVs **122**. In some examples, the computing device **200** may also use the acoustic signal data to train a machine-learning model.

[0029] As illustrated in FIG. **1**, the intelligent completion may also include an ICV control system including an ICV control device **250**. The ICV control device may be a computing device, and may be communicatively coupled to the downhole ICVs **122** by one or more ICV control cables **136**. The one or more ICV control cables may be one or more wires, one or more optical fiber cables, or another mechanism via which at least control signals from the ICV control device **250** may be transmitted to the ICVs **122**. The ICV control cable(s) **136** may be deployed into the casing **108** of the wellbore **104** in any manner described above relative to the optical fiber cable **132** of the acoustic sensing system **130**. In this example, the ICV control cable(s) **136** is affixed, such as by clamping, to the outside of the tubing string **126**. Although the ICV control cable(s) **136** and the optical fiber cable **132** of the acoustic sensing system **130** are shown to be on substantially opposite sides of the tubing string **126** for clarity in FIG. **1**, it is possible that the ICV control cable(s) **136** and the optical fiber cable **132** of the acoustic sensing system **130** may be co-located in other examples, such as in a flat pack that is clamped to the tubing string **126**.

[0030] As further illustrated in FIG. **1**, the ICV control cable(s) **136** may extend from the wellhead exit of the wellbore casing **108**, and along the surface **124** of the formation **106**, in a similar manner to the optical fiber cable **132** of the acoustic sensing system **130**. A surface ICV control cable **138** can connect the ICV control cable **136** attached to the ICV control device **250**. The ICV control device **250** may also be communicatively coupled to the computing device **200**, such that the ICV control device **250** may receive information from the computing device **200** that may be used by the ICV control device **250** to control a position of one or more of the ICVs **122**. In this manner, the acoustic sensing system **130**, the ICV control device **250**, and the computing device **200** can cooperatively form a closed loop control system. As described in more detail below, the closed loop control system may include exception handling functionality.

[0031] When the acoustic detection system **130** is a DAS system, for example, acoustic signal data may be acquired at a data rate of 1 kHz to 100 KHz depending on the optical fiber length, and more commonly between 5 kHz to 20 KHz. The data rate of a DAS system is normally significantly faster than any possible movement of an ICV and associated acoustic signals generated by the ICV movement, which can allow acoustic signals to be stacked and processed. ICVs may, in some instances, be moved in non-intuitive sequences between control points/valve positions, which may yield additional information that can be used to confirm ICV positions and system operation. For

example, acoustic signal data generated as a result of such non-intuitive movement sequences can be used to tune control algorithms or improve ICV position prediction.

[0032] The flow rate of the pressurized injection fluid **114** flowing outward through the individual ICVs **122** of the injection well **102** of FIG. **1** can be measured under known conditions (e.g., surface temperature and pressure, injection fluid temperature, etc.). For example, flow rate data associated with each of the ICVs **122** may be measured by an injection logging tool that is temporarily deployable downhole in the injection well **102** and configured to measure the flow rate of the pressurized injection fluid at various locations within the injection well **102**. The injection logging tool can be used to determine the known flow rates of the pressurized injection fluid **114** through the ICVs **122** by measuring the flow rates of the pressurized injection fluid within the completed injection well **102** at locations near each of the ICVs **122**. The location (e.g., depth) of each ICV **122** is known, whether from injection well construction data, as a result of an inspection/logging operation, or otherwise. As such, a measured pressurized injection fluid flow rate can be associated with a given ICV **122** based on the location of the injection logging tool within the injection well **102** at the time of measurement.

[0033] The acoustic sensing system **130** may be operated to record ICV acoustic signal data generated by the pressurized injection fluid flowing through the ICVs at the same time the injection logging tool is used to record pressurized injection fluid flow rates. The location of each ICV **122** within the injection well **102** is identifiable by the acoustic sensing system, thus, received acoustic signal data can be properly associated with the correct ICV **122**. Both acoustic signal data and pressurized injection fluid flow rate data for each individual ICV **122** can be measured and stored. For example, when operating the injection logging tool, the ICVs **122** may be controlled such that the pressurized injection fluid **114** flows through only one of the plurality of ICVs **122** at a time. Additionally, the positions of the ICV **122** through which the pressurized injection fluid flows may be changed in a step-wise fashion to provide a collection of acoustic signal data and pressurized injection fluid flow rate data for each ICV **122** at each of the various possible positions thereof.

[0034] The acoustic signal data recorded by the acoustic sensing system **130** can be translated to the flow rates of the pressurized injection fluid **114** flowing through the plurality of intelligent completion valves as measured by the injection logging tool. In other words, a correspondence between an acoustic signal and a pressurized injection fluid flow rate through a given ICV **122** can be established. Selected data may be stored locally, where characteristic signatures for different ICV positions and ICV position (e.g., choke setting) changes may be tracked over time. Similarly, ICV setpoint data may be used as input for injection fluid property compensation once a suitable dataset has been achieved.

[0035] The ICV acoustic signal data and the pressurized injection fluid flow rate data may be obtained from a well of current interest (e.g., the injection well **102** in which the ICVs **122** to be controlled are located), as described above. For example, an injection logging tool may be operated in the injection well **102** to record pressurized injection fluid flow rates while simultaneously recording ICV acoustic signal data using the acoustic sensing system **130**. In other examples, acoustic signal data and pressurized injection flow rate data may be obtained from a well other than a well of current interest—i.e., from a generic well. The generic well may be selected to have similar features to the injection well **102**. For example, the generic well may be located in similar environmental conditions, may be located in a similar formation, may have a similar set of ICVs, may be injected with a similar pressurized injection fluid at similar pressures, etc. In some examples, the acoustic signal data and the pressurized injection flow rate data obtained from the generic well may thus serve as baseline training data for modeling ICV operation in a well of current interest.

[0036] When processing a frame of truncated recorded ICV acoustic signal data, several filtering steps can be applied to maximize the signal-to-noise ratio and preserve the acoustic energy that corresponds to the flow of pressurized injection fluid through the ICVs. For example, a 2D median

filter with a pre-defined kernel size may first be applied. In some examples, this filtering operation may be followed by stacking of all truncated recording channels to create a single channel. Stacking the amplitude signal data by creating a single stacked channel can help to accentuate the acoustic signal of the ICV across multiple channels while canceling out noise within single channels. The ICV acoustic signal data may then be down-sampled or low pass filtered to a lower frequency. The lower frequency may be between approximately 200 Hz to approximately 2,000 Hz, with a corresponding Nyquist frequency of between approximately 100 Hz to approximately 1,000 Hz. In one non-limiting example, the ICV acoustic signal data may be down-sampled to a frequency of approximately 200 Hz with a Nyquist frequency of approximately 100 Hz by applying an anti-alias filter.

[0037] Acoustic signal data associated with the ICVs **122** and recorded by the acoustic sensing system **130** may be usable to train a machine-learning model. The training data may also include other data, such as various engineered features data. In some examples, the engineered features data may include surface conditions such as temperature or pressure, injection fluid temperature or flow rate, etc. Some of the recorded data may also be used in real time to update allocated pressurized injection fluid flow values, as the surface flow rates and pressures may fluctuate.

[0038] To further prepare the data for machine-learning model training, the truncated and signal-processed data can be converted from a time domain to a frequency domain via an integral transform, such as via the Fourier Transform. Once in the frequency domain, the number of data samples may be reduced (e.g., from 256 to 20), which can provide a smoother representation of the spectral power (i.e., energy) of the frequency spectra. This may be accomplished in a number of ways. For example, a filterbank may be applied to compute the energy contained within defined filter bank frequency bands. The filterbank is a set of narrow banded bandpass filters, which can be applied as a dot product with frequency domain spectral representation to compute the spectral power within the filterbank frequency bands. In one example, the filterbank may be a triangular filterbank. Other types of filterbanks, such as a square filterbank or a log filterbank, may also be usable. When a triangular filterbank is utilized in one example, the triangular filterbank may include filters centered at approximately 4 Hz frequency intervals (width of 7 Hz) and with a stride of approximately 3 Hz for a 4 Hz overlap between adjacent filters. Filters centered at other frequency intervals and with other strides may be utilized in other examples. Following the development (generation of engineered filterbank features) of frequency-bound filterbank energy values, the amplitudes (energy values) thereof can be converted to the decibel scale using a logarithmic transformation.

[0039] FIG. 2 is a block diagram depicting one example of a computing device **200** that is usable, in conjunction with other components, to control an ICV. While FIG. 2 shows the computing device **200** as including certain components, other examples may involve more, fewer, or different components than are shown in FIG. 2. The computing device **200** is shown to include a processor **202** that may be communicatively coupled to a memory **204** by a bus **206**. The processor **202** can include one processor or multiple processors. Non-limiting examples of the processor **202** include a Field-Programmable Gate Array (FPGA), an application specific integrated circuit (ASIC), a microprocessor, or any combination of these. Instructions **208** may be stored in the memory **204**. Output generated by the processor **202** can be presented via various mediums, including on a display device **210** that is communicatively coupled to the processor **202** by the bus **206**.

[0040] The instructions **208** are executable by the processor **202** for causing the processor **202** to perform various operations. In some examples, the instructions **208** can include processor specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, such as C, C++, C#, Java, or Python. Through the instructions **208**, the processor **202** may operate as described above to perform the various operations of the computing device **200** related to ICV control.

[0041] The memory **204** can include one memory device or multiple memory devices. The memory

204 can be non-volatile and may include any type of memory device that retains stored information when powered off. Non-limiting examples of the memory **204** include electrically erasable and programmable read-only memory (EEPROM), flash memory, or any other type of non-volatile memory. At least some of the memory device includes a non-transitory computer-readable medium from which the processor **202** can read the instructions **208**. A non-transitory computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the processor **202** the instructions **208** or other program code. Non-limiting examples of a non-transitory computer-readable medium include magnetic disk(s), memory chip(s), ROM, random-access memory (RAM), an ASIC, a configured processor, optical storage, or any other medium from which a computer processor can read the instructions **208**.

[0042] FIG. **3** is block diagram illustrating one example of a model-training application **300** that can be executed by the computing device **200**, or by a model training subsystem or another computing device that is separate from the computing device **200**, to train a machine-learning model **302**. Training the machine-learning model **302** using the model-training application **300** can transform the machine-learning model **302** from an untrained state to a trained state—i.e., to a trained machine-learning model, such as the trained machine-learning model **400** of FIG. **4**.

[0043] Training the machine-learning model can include accessing training data that may be compiled or otherwise presented in a training dataset **304**. In some examples, the training dataset **304** can be stored in a data repository that can be accessed by the model-training application **300**. The training data may include ICV that has been recorded by the acoustic sensing system **130** and is associated with the ICVs **122** in the injection well **102**. The acoustic signal data **306** may be represented in the engineered features data **310** processed by the filterbank, which may include the engineered features described above, as well as, for example, pressure data **312** (e.g., downhole pressure gauge data or surface pressure gauge data), temperature data **314** (e.g., DTS data), or surface injection flow data **316**. The model-training application **300** may also utilize the pressurized injection fluid flow rate data **308** measured by an injection logging tool while deployed downhole in the injection well **102**. The training data may also include information such as the number of perforation clusters associated with each ICV **122** within a given perforation zone of the wellbore. The training data **306-316** can be annotated and prepared for model training.

[0044] The machine-learning model **302** can be trained by causing the model-training application **300** to provide some or all of the training data **306-316** as input to the machine-learning model **302**. In an example, the machine-learning model **302** may be trained by setting the engineered features (filterbank data) as the model inputs and setting as the model targets, the pressurized injection flow rates through the ICVs as measured by, for example, an injection logging tool. The engineered features may also constitute time domain attributes derived from the downsampled or low pass filtered stacked data, wherein a root-mean-squared (RMS) amplitude may be generated. These time domain engineered features may be generated without the conversion to the frequency domain and do not require a filterbank to be applied. In addition to the RMS amplitudes computed in the time domain, attributes such a zero crossing rate may be generated as a proxy to the dominant frequency content of the data and used as inputs to the machine learning model. A soft constraint may also be applied via a physics-based model to ensure logical predictions are developed. In some examples, the machine-learning model **302** may be a neural network, such as multi-head convolutional neural network, and the machine-learning model **302** may be trained using supervised learning.

Parameters or hyperparameters may be selected to find the model parameters that optimize the fit between prediction and actual outputs. Various other fitting, estimation, or other model-training optimization techniques can be used to ensure that, upon evaluation, the predictive output of the machine-learning model **302** is accurate given the input data.

[0045] In some examples, the neural network may be used to train the machine-learning model **302** in a relative sense (versus absolute) and may utilize a flow partitioning scheme. During an injection monitoring process, the nature of the pressurized injection fluid being delivered to the injection

well **102** is known, as is the surface flow rate and the surface temperature of the pressurized injection fluid. The positions of the ICVs **122** are also known (i.e., fully open/fully closed, choked), and the pressurized injection fluid can only pass through the open ICVs **122**. In this sense, the acoustic response of the closed ICVs **122** can be set to null as inputs to the machine-learning model and the ICVs **122** that are open will account for 100% of the flow of pressurized injection fluid and can be partitioned as such (solving for the pressurized injected fluid flow through the set of ICVs as a system). Noise floor spectral amplitudes need not be known because each open ICV **122** will experience the same background noise and, therefore, the background noise will not impact flow partitioning. As such, the machine-learning model only need solve for the relationship between the acoustic signals of the open ICVs **122** and the pressurized injection fluid flow rates, while minimizing the residual between true pressurized injection fluid flow rates recorded by the injection logging tool and that which is predicted by the machine-learning model (i.e., while minimizing the loss function).

[0046] Execution of the model-training application **300** of FIG. **3** on the training data **306-316** in the training dataset **304** can output a trained machine-learning model **400** (see FIG. **4**) with an optimized set of parameters and hyperparameters for use in machine-learning inference. In the example of FIG. **4**, the trained machine-learning model **400** can be stored on the computing device **200**. Using the trained machine-learning model **400**, the computing device **200** can be used to continuously monitor a well and to generate an output when new unlabeled input data **402** in the form of new ICV acoustic signal data **404** is applied to the trained machine-learning model **400**.

[0047] The trained machine-learning model **400** may be configured to generate various types of results. For example, when the new ICV acoustic signal data **404** associated with a given ICV is applied to the trained machine-learning model **400**, the trained machine-learning model **400** can generate a result indicating a predicted flow rate of the pressurized injection fluid through the intelligent completion valve. That is, the result generated by the trained machine-learning model **400** may be a regression prediction of the flow rates of the pressurized injection fluid through each of the ICVs **122**. Each prediction may have an associated uncertainty indicating the consistency of the acoustic response across the frequencies of interest. In some examples, where multiple perforation clusters reside in a given interval/zone and are associated with a given ICV, the trained machine-learning model **400** may further generate a result indicating a predicted flow rate through each of the different perforation clusters.

[0048] In some examples, the predicted flow rate of the pressurized injection fluid through the intelligent completion valve included in the result generated by the trained machine-learning model **400** may be output to an operator or to a separate ICV control device **250**. The ICV control device **250** receiving such information may be, according to some examples, a supervisory computer system for data processing, visualization, or command and control of setpoints related to the production of hydrocarbons from a well or well system with which the ICVs associated with the notification are a part. When received by a ICV control device **250**, the result information output by the trained machine-learning model **400** may be displayed together with surface flow rates, surface pressure, flow allocation per ICV, or other information related to hydrocarbon well operations.

[0049] In some examples, the computing device **200** or the ICV control device **250** may include an automated exception handling routine whereby the computing device **200** or the ICV control device **250** can conditionally take an action. For example, the ICV control device **250** may, after receiving the result generated by the trained machine-learning model **400**, set a target flow allocation of the pressurized fluid per ICV, minimum/maximum flow rates per ICV, allowable deviations from the target flow allocation before an exception handling request is triggered, or may initially set a starting point of ICV positions to generate the target flow allocation for a given well.

[0050] In some examples, the output **406** of the trained machine-learning model **400**, which is the predicted flow rate of pressurized injection fluid through the ICV **408**, may be usable to control one or more ICVs to allocate the flow of pressurized injection fluid emitted by a well. For example, the

output **406** of the trained machine-learning model **400** and continued monitoring by an acoustic sensing system may allow the computing device **200** (or the ICV control device **250**) to make fully automated ICV position adjustments to effect flow allocation changes, which can obviate the need for an onsite operator and associated manual ICV position adjustments. Proper flow allocation can be better ensured through such automated control and, resultantly, actual flow rates through a formation can be more easily and accurately matched to and maintained at target flow rates.

[0051] FIG. **5** is a flow chart **500** representing one example of a method for controlling an ICV located downhole in a completed injection well to allocate pressurized injection fluid leaving the well. As represented in FIG. **5**, at block **502**, a processor of a computing device can receive, from an acoustic sensing system, sensor data associated with an ICV of a plurality of ICVs positionable downhole in a wellbore in a formation. In some examples, the acoustic sensing system can continuously monitor the ICVs in the wellbore and the processor may continuously receive sensor data from the acoustic sensing system. In some examples, the processor may instead receive sensor data from the acoustic sensing system at predetermined intervals. In some examples, the completed injection well may be associated with at least one completed offset producing well. In such an example, the injection well may emit pressurized injection fluid into the formation to encourage hydrocarbons located therein to move to the producing well for extraction. In some examples, the pressurized injection fluid can be water.

[0052] At block **504**, the processor can apply the sensor data to a trained machine-learning model. The trained machine-learning model may have been previously trained using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of ICVs at various flow rates and at different ICV position settings. The acoustic signal data may be recorded by the acoustic sensing system (e.g., a DAS system). In some examples, the training data may be built by operating an injection logging tool downhole in the wellbore to determine the known flow rates of the pressurized injection fluid through the plurality of ICVs while simultaneously operating the acoustic sensing system to record associated acoustic signal data.

[0053] At block **506**, the trained machine-learning model may generate a result indicating a predicted flow rate of the pressurized injection fluid through the ICV. In some examples, the machine learning model may continuously predict the flow rate of the pressurized injection fluid through the ICV, such that any deviation of the predicted flow rate of the pressurized injection fluid from a target flow rate can be identified and corrected as early as possible. In other examples, the machine learning model may predict the flow rate of the pressurized injection fluid through the ICV at predetermined intervals, which may help to conserve computing resources. In this case, the length of the predetermined intervals may be sufficiently short to ensure that any deviation of the predicted flow rate of the pressurized injection fluid from a target flow is detectable and correctable prior to having any significant impact on hydrocarbon production or before harming the formation.

[0054] At block **508**, the result can be output and can be used to adjust a position of the intelligent completion valve or a pressure of the pressurized injection fluid to control the flow rate of the pressurized injection fluid through the ICV. For example, a computing system may adjust the position of one or more of the ICVs in a well to change a flow rate allocation of the pressurized injection fluid among the ICVs. Increasing the overall flow rate of a well may not require knowledge of the peak amplitude of each ICV acoustic signal. Rather, an increase in the overall flow rate of the pressurized injection fluid leaving a well may be determined from the target flow rate and the ratio of ICV acoustic signal amplitudes (values). Alternatively, or additionally, a computing system may adjust the pressure of the pressurized injection fluid if the predicted flow rate of the pressurized injection fluid through the intelligent completion valve deviates from a target flow rate.

[0055] When the computing device or an ICV control system communicatively coupled to the computing device includes an automated exception handling routine, a method for allocating the

flow rate from an injection well may tune the ICV settings to achieve the target flow allocation per ICV within given tolerances. Execution of the exception handling routine may be triggered when the actual measured flow of the pressured fluid materially disagrees with the output of the trained machine-learning model for a given ICV position, or when the flow of the pressurized fluid deviates outside of an allowable range of target flow values. The action initiated by the exception handling routine may include specific activities tailored to the condition that triggered execution of the exception handling routine. For example, routine events like flow rates above or below target flow rate tolerances may trigger an ICV shift to reduce or increase the ICV flow area and associated flow rate through the ICV or a pressurized injection fluid pressure change, whereas a surface equipment failure may require a system re-set. In some examples, a failure to resolve an event that triggered execution of the exception handling routine may result in the generation of an alarm to one or more of the computing device, the process control system, or an operator.

[0056] According to aspects of the present disclosure, a system, a method, and a non-transitory computer-readable medium, are provided according to one or more of the following examples. As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

[0057] Example 1 is a system comprising: a plurality of intelligent completion valves positionable downhole in a wellbore in a formation; a distributed acoustic sensing system positioned in the wellbore; a processor; and a memory communicatively coupled to the processor, the memory including instructions that are executable by the processor to cause the processor to perform operations comprising: storing a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; receiving, from the distributed acoustic sensing system, sensor data associated with an intelligent completion valve of the plurality of intelligent completion valves; applying the sensor data to the trained machine-learning model to generate a result indicating a predicted flow rate of the pressurized injection fluid through the intelligent completion valve; and outputting the result that is useable to control a position of the intelligent completion valve or a pressure of the pressurized injection fluid to control the flow rate of the pressurized injection fluid through the intelligent completion valve.

[0058] Example 2 is the system of example 1, wherein the distributed acoustic sensing system includes an optical fiber cable located in the wellbore.

[0059] Example 3 is the system of example 1, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

[0060] Example 4 is the system of example 1, wherein the training data is buildable by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

[0061] Example 5 is the system of example 1, wherein the operations further comprise processing the acoustic signal data by: truncating the acoustic signal data about a location of the intelligent completion valve; filtering the acoustic signal data to maximize a signal-to-noise ratio; stacking at least some acoustic signal data truncated recording channels to create a single channel; converting the acoustic signal data from a time domain to a frequency domain by an integral transform; down-sampling or low pass filtering a frequency content of the acoustic signal data to a lower frequency; reducing a number of acoustic signal data samples by subjecting the acoustic signal data to a

filterbank; and generating engineered filterbank features having frequency-bound filterbank amplitudes, and converting the frequency-bound filterbank amplitudes to a decibel scale using a logarithmic transformation.

[0062] Example 6 is the system of example 5, wherein: the lower frequency is between approximately 200 Hz to approximately 2,000 Hz, with a corresponding Nyquist frequency of between approximately 100 Hz to approximately 1,000 Hz; and the filterbank includes filters that are centered at approximately 5 Hz to approximately 20 Hz frequency intervals with an overlap of approximately 5 Hz to approximately 20 Hz between adjacent filters.

[0063] Example 7 is the system of example 1, wherein: a flow allocation model is configured to control a flow rate of the pressurized injection fluid through each of the plurality of intelligent completion valves; and the operations further comprise automatically adjusting a position of a given intelligent completion valve to cause a flow rate of the pressurized injection fluid through the given intelligent completion valve to match a flow rate calculated for the given intelligent completion valve by the flow allocation model based on a predicted flow rate of the pressurized injection fluid through the given intelligent completion valve determined by the trained machine-learning model.

[0064] Example 8 is a method, comprising: receiving, by a processor of a computing device, from a distributed acoustic sensing system, sensor data associated with an intelligent completion valve of a plurality of intelligent completion valves positionable downhole in a wellbore in a formation; applying the sensor data to a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; generating a result indicating a predicted flow rate of pressurized injection fluid through the intelligent completion valve; and outputting the result that is used to control a position of the intelligent completion valve or a pressure of the pressurized injection fluid to control the flow rate of the pressurized injection fluid through the intelligent completion valve.

[0065] Example 9 is the method of example 8, wherein: the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

[0066] Example 10 is the method of example 8, wherein the trained machine-learning model continuously predicts the flow rate of pressurized injection fluid through the intelligent completion valve.

[0067] Example 11 is the method of example 8, wherein the training data is built by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

[0068] Example 12 is the method of example 8, further comprising processing the acoustic signal data by: truncating the acoustic signal data about a location of the intelligent completion valve; filtering the acoustic signal data to maximize a signal-to-noise ratio; stacking at least some acoustic signal data truncated recording channels to create a single channel; converting the acoustic signal data from a time domain to a frequency domain by an integral transform; down-sampling or low pass filtering a frequency of the acoustic signal data to a lower frequency; reducing a number of acoustic signal data samples by subjecting the acoustic signal data to a filterbank; and generating engineered filterbank features having frequency-bound filterbank amplitudes, and converting the frequency-bound filterbank amplitudes to a decibel scale using a logarithmic transformation.

[0069] Example 13 is the method of example 12, wherein: the lower frequency is between

approximately 200 Hz to approximately 2,000 Hz, with a Nyquist frequency of between approximately 100 Hz to approximately 1,000 Hz; and the filterbank includes filters that are centered at approximately 5 Hz to approximately 20 Hz frequency intervals with an overlap of approximately a 5 Hz to approximately 20 Hz between adjacent filters.

[0070] Example 14 is the method of example 8, wherein: a flow allocation model is configured to control a flow rate of the pressurized injection fluid through each of the plurality of intelligent completion valves; and a position of a given intelligent completion valve is automatically adjusted to cause a flow rate of the pressurized injection fluid through the given intelligent completion valve to match a flow rate calculated for the given intelligent completion valve by the flow allocation model based on a predicted flow rate of the pressurized injection fluid through the given intelligent completion valve determined by the trained machine-learning model.

[0071] Example 15 is a non-transitory computer-readable medium comprising instructions that are executable by a processor of a computing device, for causing the processor to perform operations comprising: receiving, by a processor of a computing device, from a distributed acoustic sensing system, sensor data associated with an intelligent completion valve of a plurality of intelligent completion valves positionable downhole in a wellbore in a formation; applying the sensor data to a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; generating a result indicating a predicted flow rate of pressurized injection fluid through the intelligent completion valve; and outputting the result that is useable to control a position of the intelligent completion valve or a pressure of the pressurized injection fluid to control the flow rate of the pressurized injection fluid through the intelligent completion valve.

[0072] Example 16 is the non-transitory computer-readable medium of example 15, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

[0073] Example 17 is the non-transitory computer-readable medium of example 15, wherein the training data is buildable by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

[0074] Example 18 is the non-transitory computer-readable medium of example 15, wherein the operations further comprise processing the acoustic signal data by: truncating the acoustic signal data about a location of the intelligent completion valve; filtering the acoustic signal data to maximize a signal-to-noise ratio; stacking at least some acoustic signal data truncated recording channels to create a single channel; converting the acoustic signal data from a time domain to a frequency domain by an integral transform; down-sampling or low pass filtering a frequency content of the acoustic signal data to a lower frequency; reducing a number of acoustic signal data samples by subjecting the acoustic signal data to a filterbank; and generating engineered filterbank features having frequency-bound filterbank amplitudes, and converting the frequency-bound filterbank amplitudes to a decibel scale using a logarithmic transformation.

[0075] Example 19 is the non-transitory computer-readable medium of example 18, wherein: the lower frequency is between approximately 200 Hz to approximately 2,000 Hz, with a corresponding Nyquist frequency of between approximately 100 Hz to approximately 1,000 Hz; and the filterbank includes filters that are centered at approximately 5 Hz to approximately 20 Hz

frequency intervals with an overlap of approximately 5 Hz to approximately 20 Hz between adjacent filters.

[0076] Example 20 is the non-transitory computer-readable medium of example 15, wherein: a flow allocation model is configured to control a flow rate of the pressurized injection fluid through each of the plurality of intelligent completion valves; and the operations further comprise automatically adjusting a position of a given intelligent completion valve to cause a flow rate of the pressurized injection fluid through the given intelligent completion valve to match a flow rate calculated for the given intelligent completion valve by the flow allocation model based on a predicted flow rate of the pressurized injection fluid through the given intelligent completion valve determined by the trained machine-learning model.

[0077] The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure.

Claims

1. A system comprising: a plurality of intelligent completion valves positionable downhole in a wellbore in a formation; a distributed acoustic sensing system positioned in the wellbore; a processor; and a memory communicatively coupled to the processor, the memory including instructions that are executable by the processor to cause the processor to perform operations comprising: storing a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; receiving, from the distributed acoustic sensing system, sensor data associated with an intelligent completion valve of the plurality of intelligent completion valves; applying the sensor data to the trained machine-learning model to generate a result indicating a predicted flow rate of the pressurized injection fluid through the intelligent completion valve; and outputting the result that is useable to control a position of the intelligent completion valve or a pressure of the pressurized injection fluid to control the flow rate of the pressurized injection fluid through the intelligent completion valve.
2. The system of claim 1, wherein the distributed acoustic sensing system includes an optical fiber cable located in the wellbore.
3. The system of claim 1, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.
4. The system of claim 1, wherein the training data is buildable by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.
5. The system of claim 1, wherein the operations further comprise processing the acoustic signal data by: truncating the acoustic signal data about a location of the intelligent completion valve; filtering the acoustic signal data to maximize a signal-to-noise ratio; stacking at least some acoustic signal data truncated recording channels to create a single channel; converting the acoustic signal data from a time domain to a frequency domain by an integral transform; down-sampling or low

pass filtering a frequency content of the acoustic signal data to a lower frequency; reducing a number of acoustic signal data samples by subjecting the acoustic signal data to a filterbank; and generating engineered filterbank features having frequency-bound filterbank amplitudes, and converting the frequency-bound filterbank amplitudes to a decibel scale using a logarithmic transformation.

6. The system of claim 5, wherein: the lower frequency is between approximately 200 Hz to approximately 2,000 Hz, with a corresponding Nyquist frequency of between approximately 100 Hz to approximately 1,000 Hz; and the filterbank includes filters that are centered at approximately 5 Hz to approximately 20 Hz frequency intervals with an overlap of approximately 5 Hz to approximately 20 Hz between adjacent filters.

7. The system of claim 1, wherein: a flow allocation model is configured to control a flow rate of the pressurized injection fluid through each of the plurality of intelligent completion valves; and the operations further comprise automatically adjusting a position of a given intelligent completion valve to cause a flow rate of the pressurized injection fluid through the given intelligent completion valve to match a flow rate calculated for the given intelligent completion valve by the flow allocation model based on a predicted flow rate of the pressurized injection fluid through the given intelligent completion valve determined by the trained machine-learning model.

8. A method, comprising: receiving, by a processor of a computing device, from a distributed acoustic sensing system, sensor data associated with an intelligent completion valve of a plurality of intelligent completion valves positionable downhole in a wellbore in a formation; applying the sensor data to a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; generating a result indicating a predicted flow rate of pressurized injection fluid through the intelligent completion valve; and outputting the result that is used to control a position of the intelligent completion valve or a pressure of the pressurized injection fluid to control the flow rate of the pressurized injection fluid through the intelligent completion valve.

9. The method of claim 8, wherein: the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

10. The method of claim 8, wherein the trained machine-learning model continuously predicts the flow rate of pressurized injection fluid through the intelligent completion valve.

11. The method of claim 8, wherein the training data is built by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

12. The method of claim 8, further comprising processing the acoustic signal data by: truncating the acoustic signal data about a location of the intelligent completion valve; filtering the acoustic signal data to maximize a signal-to-noise ratio; stacking at least some acoustic signal data truncated recording channels to create a single channel; converting the acoustic signal data from a time domain to a frequency domain by an integral transform; down-sampling or low pass filtering a frequency of the acoustic signal data to a lower frequency; reducing a number of acoustic signal data samples by subjecting the acoustic signal data to a filterbank; and generating engineered filterbank features having frequency-bound filterbank amplitudes, and converting the frequency-bound filterbank amplitudes to a decibel scale using a logarithmic transformation.

13. The method of claim 12, wherein: the lower frequency is between approximately 200 Hz to

approximately 2,000 Hz, with a Nyquist frequency of between approximately 100 Hz to approximately 1,000 Hz; and the filterbank includes filters that are centered at approximately 5 Hz to approximately 20 Hz frequency intervals with an overlap of approximately a 5 Hz to approximately 20 Hz between adjacent filters.

14. The method of claim 8, wherein: a flow allocation model is configured to control a flow rate of the pressurized injection fluid through each of the plurality of intelligent completion valves; and a position of a given intelligent completion valve is automatically adjusted to cause a flow rate of the pressurized injection fluid through the given intelligent completion valve to match a flow rate calculated for the given intelligent completion valve by the flow allocation model based on a predicted flow rate of the pressurized injection fluid through the given intelligent completion valve determined by the trained machine-learning model.

15. A non-transitory computer-readable medium comprising instructions that are executable by a processor of a computing device, for causing the processor to perform operations comprising: receiving, by a processor of a computing device, from a distributed acoustic sensing system, sensor data associated with an intelligent completion valve of a plurality of intelligent completion valves positionable downhole in a wellbore in a formation; applying the sensor data to a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; generating a result indicating a predicted flow rate of pressurized injection fluid through the intelligent completion valve; and outputting the result that is useable to control a position of the intelligent completion valve or a pressure of the pressurized injection fluid to control the flow rate of the pressurized injection fluid through the intelligent completion valve.

16. The non-transitory computer-readable medium of claim 15, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

17. The non-transitory computer-readable medium of claim 15, wherein the training data is buildable by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

18. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise processing the acoustic signal data by: truncating the acoustic signal data about a location of the intelligent completion valve; filtering the acoustic signal data to maximize a signal-to-noise ratio; stacking at least some acoustic signal data truncated recording channels to create a single channel; converting the acoustic signal data from a time domain to a frequency domain by an integral transform; down-sampling or low pass filtering a frequency content of the acoustic signal data to a lower frequency; reducing a number of acoustic signal data samples by subjecting the acoustic signal data to a filterbank; and generating engineered filterbank features having frequency-bound filterbank amplitudes, and converting the frequency-bound filterbank amplitudes to a decibel scale using a logarithmic transformation.

19. The non-transitory computer-readable medium of claim 18, wherein: the lower frequency is between approximately 200 Hz to approximately 2,000 Hz, with a corresponding Nyquist frequency of between approximately 100 Hz to approximately 1,000 Hz; and the filterbank includes filters that are centered at approximately 5 Hz to approximately 20 Hz frequency intervals with an overlap of approximately 5 Hz to approximately 20 Hz between adjacent filters.

20. The non-transitory computer-readable medium of claim 15, wherein: a flow allocation model is

configured to control a flow rate of the pressurized injection fluid through each of the plurality of intelligent completion valves; and the operations further comprise automatically adjusting a position of a given intelligent completion valve to cause a flow rate of the pressurized injection fluid through the given intelligent completion valve to match a flow rate calculated for the given intelligent completion valve by the flow allocation model based on a predicted flow rate of the pressurized injection fluid through the given intelligent completion valve determined by the trained machine-learning model.
